

Theorem (A)

Let V be a finite dimensional vector space with $V \neq \{0\}$
and $V = \text{Span}(S)$, S is finite.

Then there is a subset $B \subseteq S$, which is a basis of V . That is, every spanning set of V contains a basis.

Proof Let $S = \{v_1, \dots, v_n\}$.

(Algorithm) We may assume that $v_i \neq 0$ for all i .

Step 1 Check if S is linearly independent. If it is, then
it is a basis of V . Choose $B = \{v_1, \dots, v_n\}$.
Otherwise, proceed to Step 2

Step 2 Find a $j \in \{1, 2, \dots, n\}$ such that
 $\text{Span}\{v_1, v_2, \dots, v_{j-1}, v_{j+1}, \dots, v_n\} = \text{Span}(S) = V$.

Replace S by $\{v_1, v_2, \dots, v_{j-1}, v_{j+1}, \dots, v_n\}$ and apply
Step 1 □

Theorem B Every linearly independent subset of a finite dimensional vector space can be extended to a basis.

Proof Let $V = \text{Span}\{v_1, \dots, v_n\}$ and $S = \{w_1, w_2, \dots, w_m\}$ be a linearly independent subset of V .

Step 1 Check if $V = \text{Span}(S)$. If it is then S is a basis. Otherwise, Proceed to Step 2.

Step 2 pick a vector $v \in V$ with $v \notin \text{Span}(S)$. Include v in the set S . Observe that this modified set S is also a linearly independent set.

Now perform Step 1.

This process terminates because the modified set S can have at most n number of linearly independent elements. $\square$

Theorem C Any two bases of a finite dimensional vector space has the same number of elements.

Proof Let $B_1 = \{u_1, u_2, \dots, u_m\}$ and $B_2 = \{v_1, v_2, \dots, v_n\}$

be two sets of bases of a finite dimensional vector space V . Since $V = \text{Span}\{v_1, \dots, v_n\}$ and B_1 is a set consisting of m linearly independent vectors, by a theorem proved earlier

$$m \leq n$$

Similarly, $V = \text{Span}\{u_1, u_2, \dots, u_m\}$ and that B_2 is a set consisting of n linearly independent vectors implies,

$$n \leq m$$

Thus $n = m$. □

Dimension: Let V be a finite dimensional vector space. If B is a basis of V consisting of n elements then dimension of V is defined to be n : Notation: $\text{Dim}_F(V) = n$

Observations

① Every finite dimensional vector space has a basis Theorem (A)

② Since F^n has the basis $\left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$ consisting on

n elements, $\text{Dim}_F(F^n) = n$.

③ Let V be a Vector Space with $\text{Dim}_F(V) = n$
and W be a Subspace of V .
Then, $\text{Dim}_F(W) \leq \text{Dim}_F(V)$. Moreover,

$\text{Dim}_F(W) = \text{Dim}_F(V)$ if and only if $W = V$.

Proof We have already proved that W has to be finite dimensional. We just saw that W must have a basis. Since $V = \text{Span}\{v_1, \dots, v_n\}$, the number of elements in any basis of W must be $\leq n$.

If $V = W$ then, clearly, $\text{Dim}_F(V) = \text{Dim}_F(W)$.

Now Suppose that $\text{Dim}_F(V) = \text{Dim}_F(W)$.

Let $V = \text{Span}(B_1)$ and $W = \text{Span}(B_2)$, where B_1 and B_2 are bases of V and W respectively.

Note that $B_1 \neq B_2$ has the same number of elements

If $W \subsetneq V$ then pick $v \in V$ and $v \notin W$
and observe that $B_2 \cup \{v\}$ is a linearly independent set. This is a Contradiction to a theorem proved earlier which states that the number of linearly independent vectors must be at most the number of any spanning set of vectors.

Linear Transformation

A map $T: V \rightarrow W$, between $\mathbb{F}$ -vector spaces V and W is called a linear transformation if it satisfies the following conditions:

- (A) $T(v_1 + v_2) = Tv_1 + Tv_2$; $v_1, v_2 \in V$
- (B) $T(\alpha v) = \alpha T(v)$, $\alpha \in \mathbb{F}$ and $v \in V$

Example

① Let A be an $m \times n$ matrix with entries in $\mathbb{R}$. Then $A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfies the conditions ① and ②

* $A(x + y) = Ax + Ay$, where Ax and Ay are matrix multiplications.

$$* A(\alpha x) = \alpha Ax.$$

② The map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a^2 \\ b^2 \end{pmatrix}$$

is not a linear transformation $\left\{ \begin{array}{l} \text{① does not hold!} \end{array} \right.$

Let V, W be $\mathbb{F}$ -vector spaces.

Definitions

① A linear transformation $T: V \rightarrow W$ is also called a homomorphism (or simply, morphism) of vector spaces

② A one to one linear transformation $T: V \rightarrow W$ (injective)

is also called a monomorphism of vector spaces

③ A onto linear transformation $T: V \rightarrow W$ is (surjective)

also called a epimorphism of vector spaces

④ A bijective linear transformation $T: V \rightarrow W$ is also called an isomorphism of vector spaces.

In which case V and W are called isomorphic vector spaces; $V \cong W$ (equivalent spaces)

If $T: V \rightarrow W$ is an isomorphism then for all practical purposes, V and W are the same vector space

Theorem Let V be a finite dimensional vector space with $\dim_{\mathbb{F}}(V) = n$. Then $V \cong \mathbb{F}^n$.

Proof Let $\{v_1, v_2, \dots, v_n\}$ be a basis of V and $\{e_1, e_2, \dots, e_n\}$ be the standard basis of $\mathbb{F}^n$.

$$e_i := \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

1 in i th position.

Let $T: V \rightarrow \mathbb{F}^n$ be the map defined by

$$T(\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n) = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n$$

H.W Check that T is a linear transformation.

If $T(\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n) = 0$ then

$$\alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n = 0$$

$$\Rightarrow \alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_n = 0.$$

Therefore T must be injective.

For any element $\beta_1 e_1 + \beta_2 e_2 + \dots + \beta_n e_n \in \mathbb{F}^n$, we observe that

$$\begin{aligned} T(\beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n) \\ = \beta_1 e_1 + \beta_2 e_2 + \dots + \beta_n e_n. \end{aligned}$$

Therefore T is surjective

Hence $V \cong \mathbb{F}^n$

